🍽️ Restaurant Bill Calculator (GUI)

Guided Lab – Tkinter Basics

📘 Course Alignment

- **Course:** Introduction to Python Programming
- **Module:** GUI Programming with Tkinter
- **Estimated Time:** 90–120 minutes

🎯 Learning Outcomes

By completing this lab, students will be able to:

- Create a basic GUI using Tkinter
- Use `Label`, `Entry`, `Button`, and `Text` widgets
- Capture and validate user input
- Perform calculations using variables
- Display formatted output in a GUI
- Convert a console program into a GUI application

🧪 Lab Scenario

You are designing a **Restaurant Bill Calculator** that allows users to enter item prices and displays a receipt showing:

- Each item price
- Subtotal
- Sales tax (6.75%)
- Tip (20%)
- Final total

All interaction must occur through a **graphical user interface (GUI)**.

🧩 Part 1: Starter Code (Do NOT Modify Yet)

```
import tkinter as tk
from tkinter import messagebox

# Create main window
window = tk.Tk()
window.title("Restaurant Bill Calculator")
window.geometry("350x400")

# Run window
window.mainloop()
```

☑ If a blank window appears, continue.

🧩 Part 2: Add Global Variables

Add these **above** `window.mainloop()`:

```
subtotal = 0
item_number = 1
```

✔ These track the running total and item count.

🧩 Part 3: Create the GUI Layout

Add this **before** `window.mainloop()`:

```
# Label
tk.Label(window, text="Enter Item Price").pack(pady=5)

# Entry box
price_entry = tk.Entry(window)
price_entry.pack(pady=5)

# Buttons
add_button = tk.Button(window, text="Add Item")
add_button.pack(pady=5)

calc_button = tk.Button(window, text="Calculate Total")
```

```
calc_button.pack(pady=5)

# Receipt display
receipt_box = tk.Text(window, width=40, height=15)
receipt_box.pack(pady=10)
```

☑ Run to verify layout appears correctly.

🧩 Part 4: Create the `add_item()` Function

Add this **above the GUI layout**:

```
def add_item():
    global subtotal, item_number

    try:
        # Get price from entry
        price = float(price_entry.get())

        # Add to subtotal
        subtotal += price

        # Display item in receipt
        receipt_box.insert(tk.END, f"Item {item_number}: ${price:.2f}\n")

        # Increase item number
        item_number += 1

        # Clear entry box
        price_entry.delete(0, tk.END)

    except ValueError:
        messagebox.showerror("Invalid Input", "Please enter a valid number.")
```

🧩 Part 5: Create the `calculate_total()` Function

```
def calculate_total():
    tax = subtotal * 0.0675
    tip = subtotal * 0.20
    total = subtotal + tax + tip

    receipt_box.insert(tk.END, "----------------------\n")
    receipt_box.insert(tk.END, f"Subtotal: ${subtotal:.2f}\n")
    receipt_box.insert(tk.END, f"Tax: ${tax:.2f}\n")
    receipt_box.insert(tk.END, f"Tip: ${tip:.2f}\n")
    receipt_box.insert(tk.END, f"Final Total: ${total:.2f}\n")
```

🧩 Part 6: Connect Buttons to Functions

Update button code:

```
add_button.config(command=add_item)
calc_button.config(command=calculate_total)
```

☑ Final Complete Program

```
import tkinter as tk
from tkinter import messagebox

# Create main window
window = tk.Tk()
window.title("Restaurant Bill Calculator")
window.geometry("350x400")

# Global variables
subtotal = 0
item_number = 1

# Functions
def add_item():
    global subtotal, item_number
    try:
        price = float(price_entry.get())
        subtotal += price
        receipt_box.insert(tk.END, f"Item {item_number}: ${price:.2f}\n")
        item_number += 1
        price_entry.delete(0, tk.END)
    except ValueError:
        messagebox.showerror("Invalid Input", "Please enter a valid number.")

def calculate_total():
    tax = subtotal * 0.0675
    tip = subtotal * 0.20
    total = subtotal + tax + tip

    receipt_box.insert(tk.END, "----------------------\n")
    receipt_box.insert(tk.END, f"Subtotal: ${subtotal:.2f}\n")
    receipt_box.insert(tk.END, f"Tax: ${tax:.2f}\n")
    receipt_box.insert(tk.END, f"Tip: ${tip:.2f}\n")
    receipt_box.insert(tk.END, f"Final Total: ${total:.2f}\n")

# GUI Layout
tk.Label(window, text="Enter Item Price").pack(pady=5)

price_entry = tk.Entry(window)
price_entry.pack(pady=5)

add_button = tk.Button(window, text="Add Item")
```

```
add_button.pack(pady=5)

calc_button = tk.Button(window, text="Calculate Total")
calc_button.pack(pady=5)

receipt_box = tk.Text(window, width=40, height=15)
receipt_box.pack(pady=10)

# Connect buttons
add_button.config(command=add_item)
calc_button.config(command=calculate_total)

# Run window
window.mainloop()
```

🖥️ Sample Output (GUI Receipt)

```
Item 1: $12.50
Item 2: $8.25
----------------------
Subtotal: $20.75
Tax: $1.40
Tip: $4.44
Final Total: $26.59
```